Invariant encoding. Trace-preserving dynamics.

Nicholas Shane Kouns · E47 / KKP-R · 18 October 2026

125 carrier states → 47 invariant states

7 data qubits + 1 syndrome ancilla · 36 / 36 checks passed

01 SELECT THE INVARIANT SPACE

$$H = V_2 \otimes V_2 \otimes V_2 \qquad \dim H = 5^3 = 125$$

$$C = J^2 \qquad K = (C - 6I)(C - 30I)$$

$$E_{47} = \ker K = 5V_2 \oplus 2V_5 \qquad \dim E_{47} = 25 + 22 = 47$$

Proof: K vanishes exactly at Casimir eigenvalues 6 and 30; their spin-sector dimensions are 25 and 22.

02 ENCODE AND MEASURE

$$|j, j_{12}, m\rangle \leftrightarrow \text{packed indices } 0\ldots124;\ 125\ldots127 \text{ unused}$$

$$W = V^\dagger \oplus I_3 \qquad P = \Sigma_{k\in S} |k\rangle\langle k| \qquad |S| = 47$$

$$\Pr(\text{syndrome} = 1) = \mathrm{Tr}(P\, W\rho W^\dagger)$$

V contains the joint eigenvectors of C, C_{12} and Jz. Apply W, flip the ancilla on S, then measure. All 128 packed inputs were checked.

03 PRESERVE TRACE AND THE KERNEL

$$\gamma_k = 1 - [(\lambda_k - 6)(\lambda_k - 30)]^2 / 99144$$

$$A = \mathrm{diag}(\gamma_0,\ldots,\gamma_{124}, 0, 0, 0) \qquad \tau = |\alpha\rangle\langle\alpha|,\ \alpha \in E_{47}$$

$$E(\rho) = A\rho A^\dagger + \mathrm{Tr}[(I - A^\dagger A)\rho]\,\tau$$

Proof: $B_k = \sqrt{(1 - \gamma_k^2)}|\alpha\rangle\langle k|$ gives $A^\dagger A + \Sigma_k B_k^\dagger B_k = I$. Therefore E is completely positive and trace preserving; $E(\rho) = \rho$ on E_{47}.

04 CONTRACT THE COMPLEMENT

$$Q = I - P \qquad r = 15/17 \qquad \|AQ\|_2 = r$$

$$\mathrm{Tr}[Q\, E^n(\rho)] \le r^{2n}\, \mathrm{Tr}(Q\rho)$$

$$\lim_{n\to\infty} E^n(\rho) = P\rho P + \mathrm{Tr}(Q\rho)\,\tau$$

Selected information stays fixed. Complement population transfers to τ; coherence between the two spaces decays at most as r^n.

EXECUTED EVIDENCE / PYTHON + QISKIT

Synthesized circuit error: 2.64×10^{-13} | Channel agreement: 8.55×10^{-17}

Double-precision simulation; no quantum hardware run. The pure reset target is an explicit choice.

S = {10,...,34} ∪ {90,...,111}. All equations after W use packed coordinates.